



PyConES25

Del Papel al Conocimiento: Creando un Sistema RAG Multimodal con Docling



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PyConES 25



Introducción a Docling

~10 minutos

- ¿Que es Docling?
- Motivación y Ejemplos



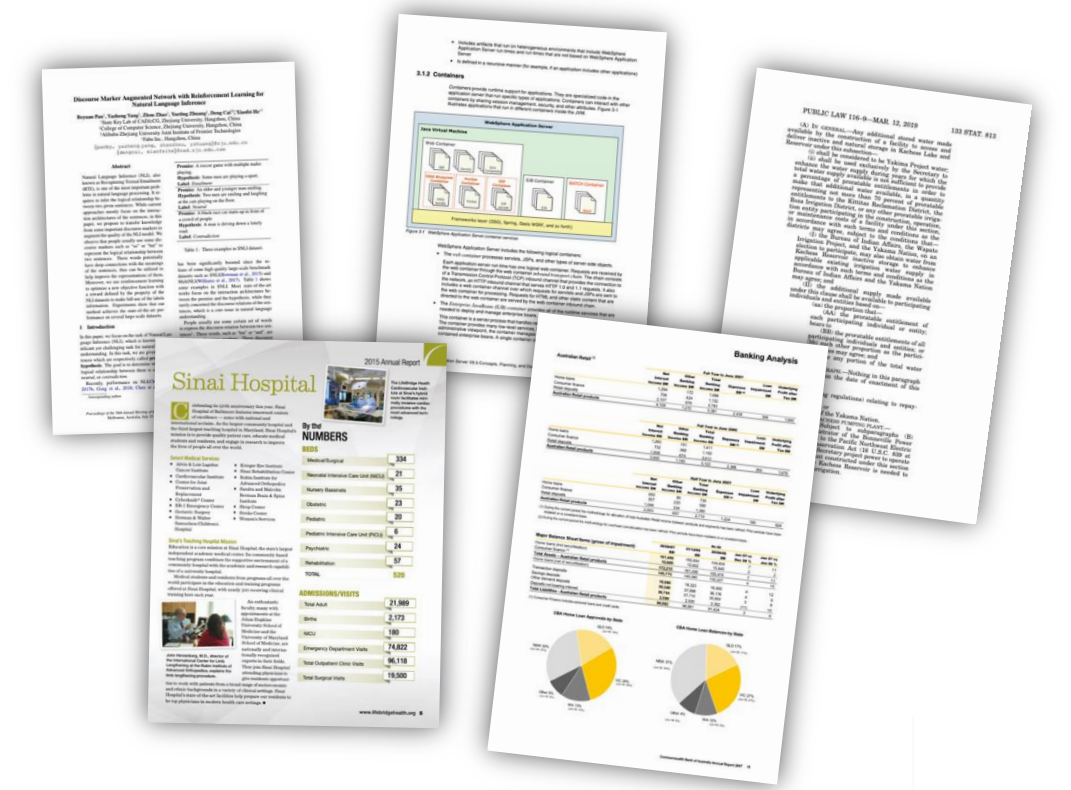
Docling Hands-On

25-30 min c/u

- Lab 1: Conversión de Documentos
- Lab 2: Chunking Avanzado con Docling
- Lab 3: RAG Multimodal con Docling

Introducing Docling

- 📁 Parsing of multiple document formats incl. PDF, DOCX, XLSX, HTML, images, and more
- 📄 Advanced PDF understanding incl. page layout, reading order, table structure, code, formulas, image classification, ...
- 🧬 Unified, expressive DoclingDocument representation format
 - ↪ Various export formats (Markdown, HTML, JSON)
- 🔒 Local execution for sensitive data and air-gapped environments
- 🤖 Many plug-and-play ecosystem integrations
- 🔍 Extensive OCR support for scanned PDFs and images
- 🧠 Support of Visual Language Models
- 🗣️ Support for Audio with Automatic Speech Recognition models
- 💻 Simple and convenient CLI



```
pip install docling

# a single document to markdown
docling https://arxiv.org/pdf/2408.09869.pdf

# a folder of documents to markdown and json
docling --to json --to md ./inputs/
```



🚀 Use it, ★ star it,
spread the word about
Docling!

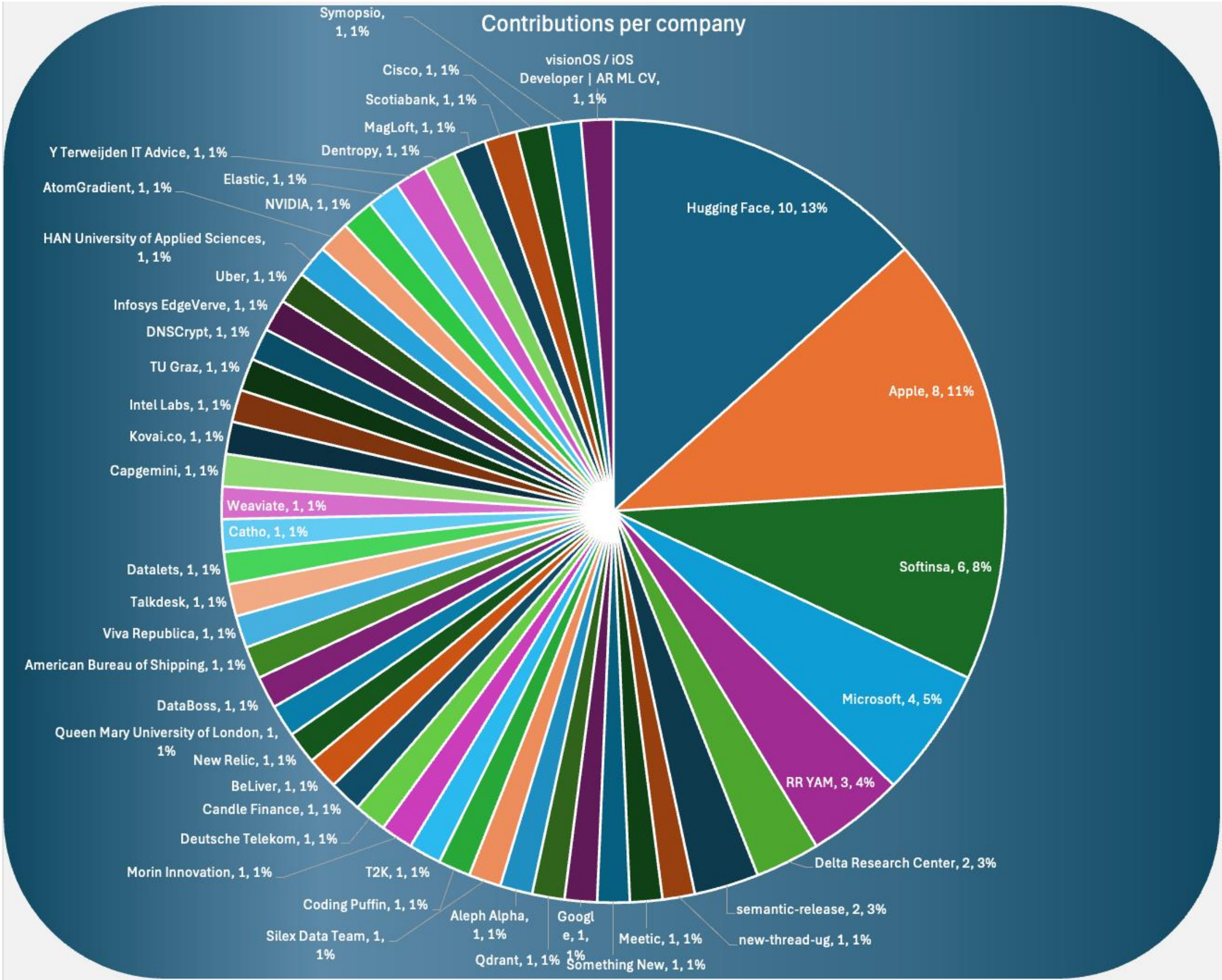
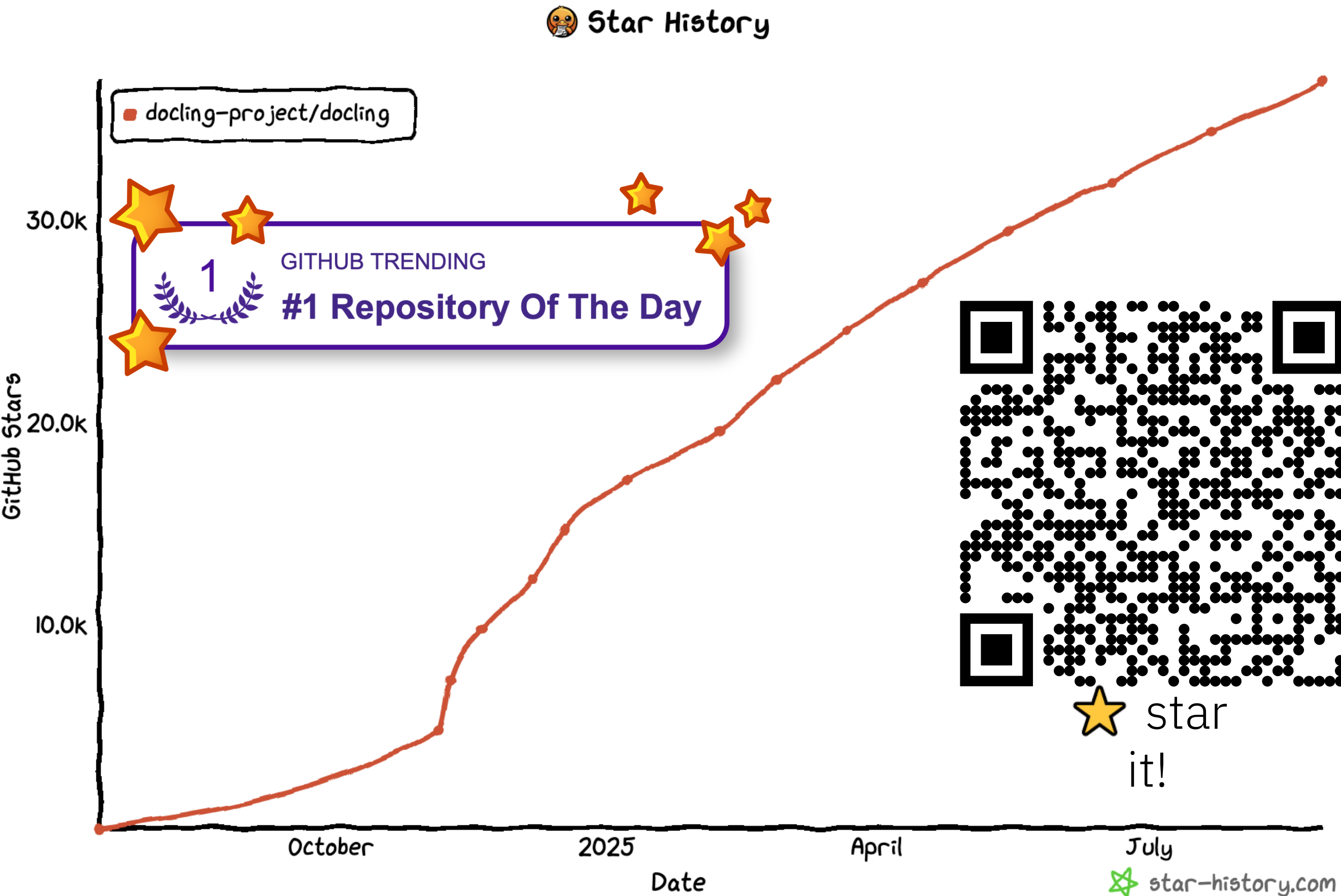
Community Adoption

A rising star in the open-source space

- ★ 39k+ GitHub stars
- 🐍 1M+ downloads/month from [PyPi](#)

Used by 2.2k

 + 2,149



Ecosystem

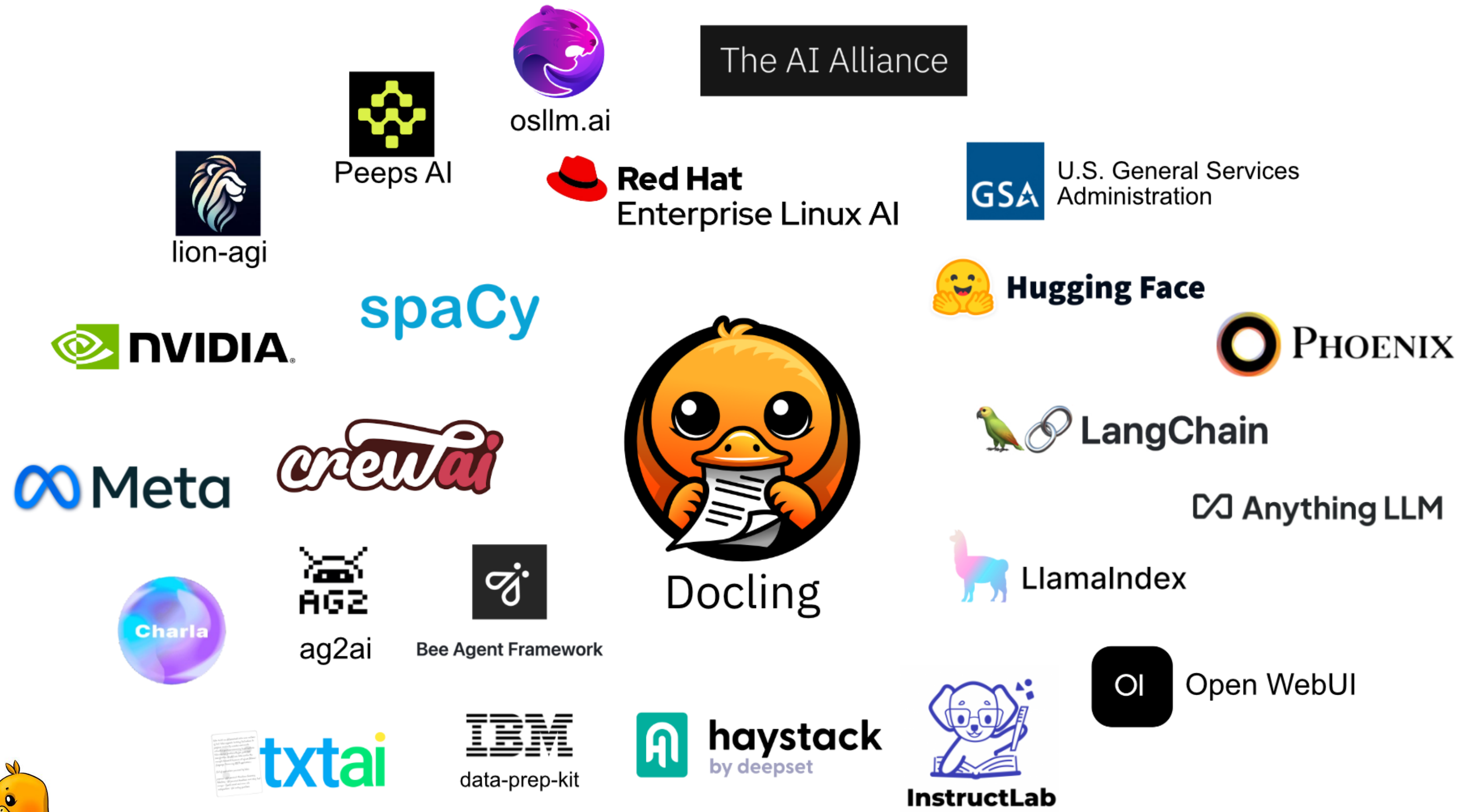
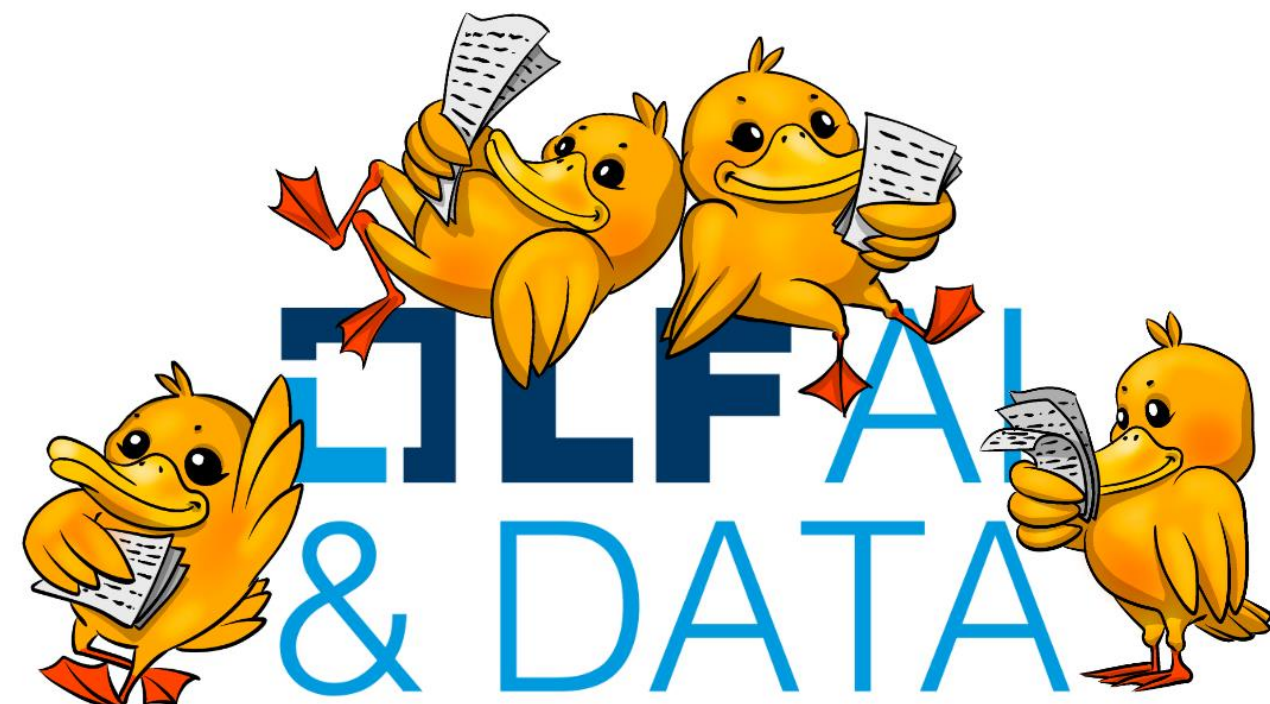
RAG Frameworks

Agentic systems

Dataset generation

Vector databases

Client solutions



Demo

Recovering structured content from PDF

with low-level PDF parsers

! undesired
! page headers

KDD '22, August 14–18, 2022, Washington, DC, USA Birgit Pfitzmann, Christoph Auer, Michele Dolfi, Ahmed S. Nassar, and Peter Staar

Table 1: DocLayNet dataset overview. Along with the frequency of each class label, we present the relative occurrence (as % of row “Total”) in the train, test and validation sets. The inter-annotator agreement is computed as the mAP@0.5-0.95 metric between pairwise annotations from the triple-annotated pages, from which we obtain accuracy ranges.

class label	Count	% of Total			triple inter-annotator mAP @ 0.5-0.95 (%)									
		Train	Test	Val	All	Fin	Man	Sci	Law	Pat	Ten			
Caption	22524	2.04	1.77	2.32	84-89	40-61	86-92	94-99	95-99	69-78	n/a			
Footnote	6318	0.60	0.31	0.58	83-91	n/a	100	62-88	85-94	n/a	82-97			
Formula	25027	2.25	1.90	2.96	83-85	n/a	n/a	84-87	86-96	n/a	n/a			
List-item	185660	17.19	13.34	15.82	87-88	74-83	90-92	97-97	81-85	75-88	93-95			
Page-footer	70878	6.51	5.58	6.00	93-94	88-90	95-96	100	92-97	100	96-98			
Page-header	58022	5.10	6.70	5.06	85-89	66-76	90-94	98-100	91-92	97-99	81-86			
Picture	45976	4.21	2.78	5.31	69-71	56-59	82-86	69-82	80-95	66-71	59-76			
Section-header	142884	12.60	15.77	12.85	83-84	76-81	90-92	94-95	87-94	69-73	78-86			
Table	34733	3.20	2.27	3.60	77-81	75-80	83-86	98-99	58-80	79-84	70-85			
Text	510377	45.82	49.28	45.00	84-86	81-86	88-93	89-93	87-92	71-79	87-95			
Title	5071	0.47	0.30	0.50	60-72	24-63	50-63	94-100	82-96	68-79	24-56			
Total	1107470	941123	99816	66531	82-83	71-74	79-81	89-94	86-91	71-76	68-85			

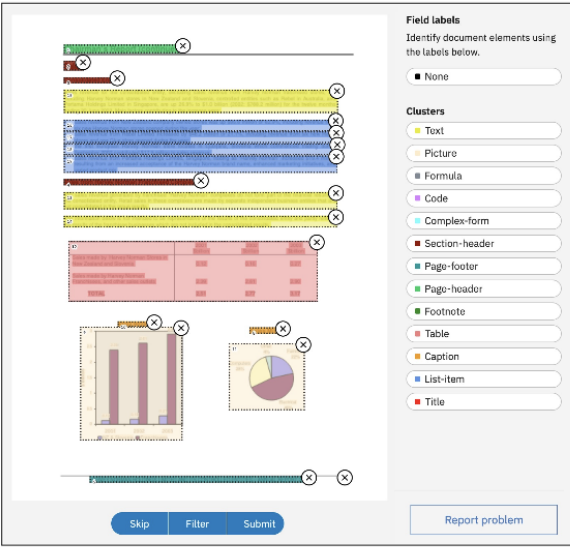


Figure 3: Corpus Conversion Service annotation user interface. The PDF page is shown in the background, with overlaid text-cells (in darker shades). The annotation boxes can be drawn by dragging a rectangle over each segment with the respective label from the palette on the right.

we distributed the annotation workload and performed continuous quality controls. Phase one and two required a small team of experts only. For phases three and four, a group of 40 dedicated annotators were assembled and supervised.

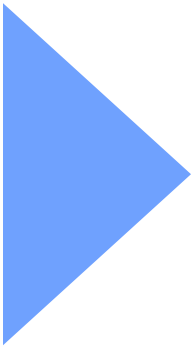
Phase 1: Data selection and preparation. Our inclusion criteria for documents were described in Section 3. A large effort went into ensuring that all documents are free to use. The data sources

include publication repositories such as arXiv³, government offices, company websites as well as data directory services for financial reports and patents. Scanned documents were excluded wherever possible because they can be rotated or skewed. This would not allow us to perform annotation with rectangular bounding-boxes and therefore complicate the annotation process.

Preparation work included uploading and parsing the sourced PDF documents in the Corpus Conversion Service (CCS) [22], a cloud-native platform which provides a visual annotation interface and allows for dataset inspection and analysis. The annotation interface of CCS is shown in Figure 3. The desired balance of pages between the different document categories was achieved by selective subsampling of pages with certain desired properties. For example, we made sure to include the title page of each document and bias the remaining page selection to those with figures or tables. The latter was achieved by leveraging pre-trained object detection models from PubLayNet, which helped us estimate how many figures and tables a given page contains.

Phase 2: Label selection and guideline. We reviewed the collected documents and identified the most common structural features they exhibit. This was achieved by identifying recurrent layout elements and lead us to the definition of 11 distinct class labels. These 11 class labels are *Caption*, *Footnote*, *Formula*, *List-item*, *Page-footer*, *Page-header*, *Picture*, *Section-header*, *Table*, *Text*, and *Title*. Critical factors that were considered for the choice of these class labels were (1) the overall occurrence of the label, (2) the specificity of the label, (3) recognisability on a single page (i.e. no need for context from previous or next page) and (4) overall coverage of the page. Specificity ensures that the choice of label is not ambiguous, while coverage ensures that all meaningful items on a page can be annotated. We refrained from class labels that are very specific to a document category, such as *Abstract* in the *Scientific Articles* category. We also avoided class labels that are tightly linked to the semantics of the text. Labels such as *Author* and *Affiliation*, as seen in DocBank, are often only distinguishable by discriminating on

³<https://arxiv.org/>



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% of Total

triple inter-annotator mAP @ 0.5-0.95 (%)

[...]
Count
22524
6318
25027
185660
70878
58022
45976
142884
34733
510377
5071
1107470

! Tables not
! understood

! Image content
! missing

[...]

include publication repositories such as arXiv³, government offices, company websites as well as data directory services for financial reports and patents. Scanned documents were excluded wherever possible because they can be rotated or skewed. This would not allow us to perform annotation with rectangular bounding-boxes and therefore complicate the annotation process.

[...]
! Multi-column often
! breaks order

! Line wraps not
! understood

✓ Very fast and cheap

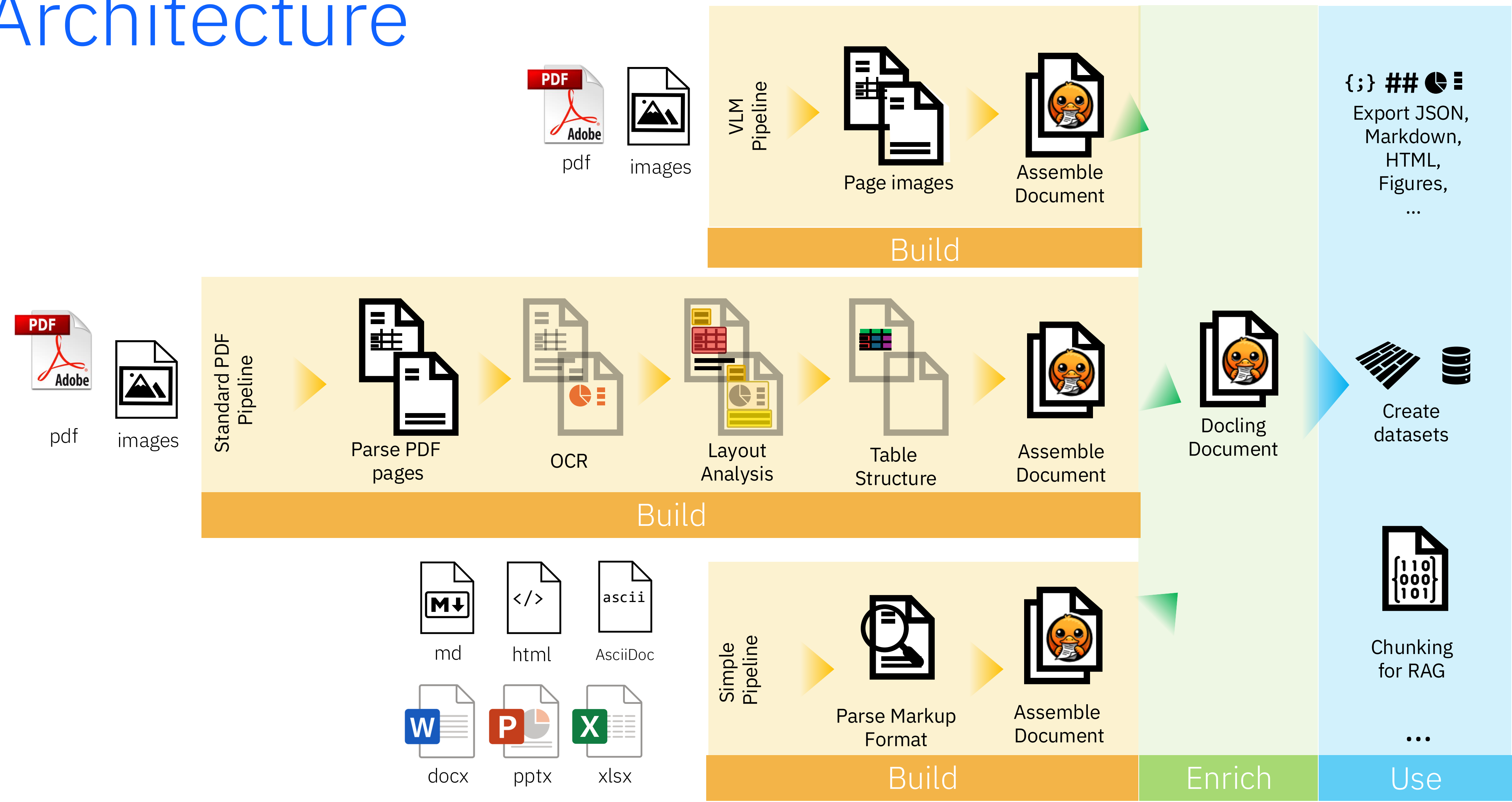
✗ Incomplete

✗ Loss of structure

✗ Noisy

➡ Unfit for most use cases

Design and Architecture



A new vlm generation

Ultra-small, built on specialized, llm-friendly tokens

SmolDocling

Single VLM Model for
OCR/layout/table/chart

Mar 17



Trending on 🤖 this week

Models

ds4sd/SmolDocling-256M-preview

Updated 1 day ago • 📄 27.9k • ❤️ 847

mistralai/Mistral-Small-3.1-24B-Ins...

Updated 2 days ago • 📄 60.4k • ❤️ 924

manycore-research/SpatialLM-Llama-1B

Updated 4 days ago • 📄 2.38k • ❤️ 588

sesame/csm-1b

Updated 8 days ago • 📄 32k • ❤️ 1.58k

deepseek-ai/DeepSeek-V3-0324

Updated about 6 hours ago • ❤️ 471

[Browse 1M+ models](#)



Total downloads
+950k

Sep 17

Granite Docling

Enhanced stability
Production quality
Multi-lingual

Like it ❤️



[granite-docling-258M](#)



Demo

<https://huggingface.co/spaces/ibm-granite/granite-docling-258m-demo>

Why it works – and why people love it



Fig. 3. OTSL description of table structure: A - table example; B - graphical representation of table structure; C - mapping structure on a grid; D - OTSL structure encoding; E - explanation on cell encoding

1 - simple cells: "C"
2 - horizontal merges: "C", "L"
3 - vertical merges: "C", "U"
4 - 2d merges: "C", "L", "U", "X"

Table 2. TSR and cell detection results compared between OTSL and HTML on the PubTabNet [22], FinTabNet [21] and PubTables-1M [14] data sets using TableFormer [9] (with enc=6, dec=6, heads=8).

Data set	Language	TEDs			mAP(0.75)	Inference time (secs)
		simple	complex	all		
PubTabNet	OTSL	0.965	0.934	0.955	0.88	2.73
	HTML	0.969	0.927	0.955	0.857	5.39
FinTabNet	OTSL	0.955	0.961	0.959	0.862	1.85
	HTML	0.917	0.922	0.92	0.722	3.26
PubTables-1M	OTSL	0.987	0.964	0.977	0.896	1.79
	HTML	0.983	0.944	0.966	0.889	3.26

5.3 Qualitative Results

The OTSL vocabulary is comprised of the following tokens:

- "C" cell - a new table cell that either has or does not have cell content
- "L" cell - left-looking cell, merging with the left neighbor cell to create a span
- "U" cell - up-looking cell, merging with the upper neighbor cell to create a span
- "X" cell - cross cell, to merge with both left and upper neighbor cells
- "NL" - new-line, switch to the next row.

A notable attribute of OTSL is that it has the capability of achieving lossless

Fig. 3. OTSL description of table structure: A - table example; B - graphical representation of table structure; C - mapping structure on a grid; D - OTSL structure encoding; E - explanation on cell encoding

OTSL structure encoding:

```
<picture><loc_102><loc_85><loc_416><loc_160>
<flow_chart>
<caption>
</caption>
</picture>
```

OTSL structure encoding:

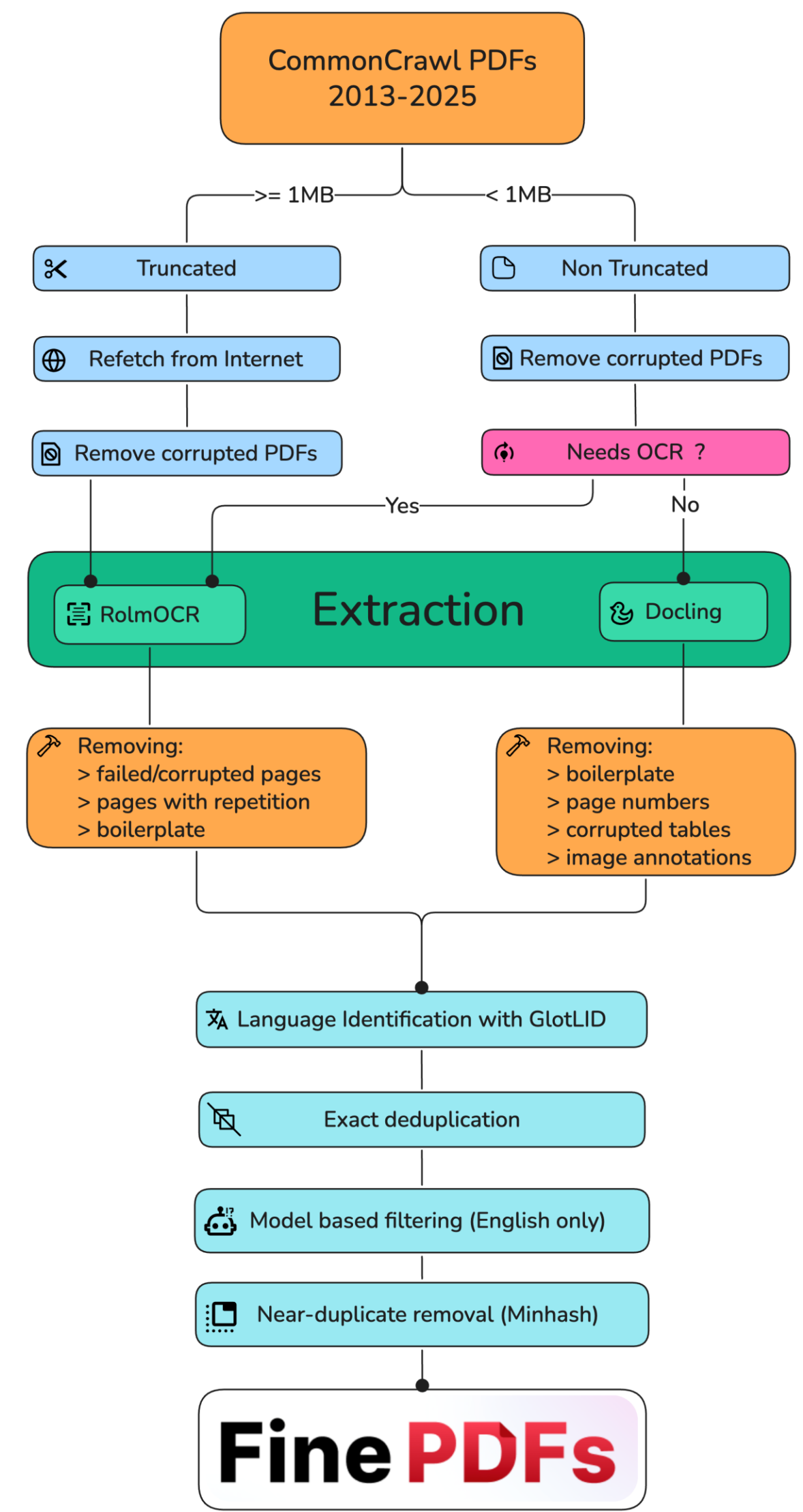
```
<otsl><loc_63><loc_78><loc_436><loc_158>
<ched>Data set <ched>Language <ched>TEDs <lcel>
<ecel> <ecel> <ched>simple <ched>
<rhed>PubTabNet <rhed>OTSL <fcel>0.965 <fcel>0
<ucel> <rhed>HTML <fcel>0.969 <fcel>0
...
<caption>Table 2. TSR and cell detection results compared betw
[14] data sets using TableFormer [9] (with enc=6, dec=6, heads=
</otsl>
```

OTSL structure encoding:

```
<section_header_level_1><loc_58><loc_343><loc_209><loc_
<text><loc_66><loc_360><loc_358><loc_368>The OTSL voca
<unordered_list>
<list_item><loc_53><loc_377><loc_439><loc_385>- "C" cell -
<list_item><loc_53><loc_387><loc_439><loc_403>- "L" cell -
<list_item><loc_53><loc_405><loc_439><loc_421>- "U" cell -
<list_item><loc_53><loc_423><loc_409><loc_431>- "X" cell -
<list_item><loc_53><loc_433><loc_269><loc_441>- "NL" - ne
</unordered_list>
```

- OCR & Bounding Boxes** – Accurate text extraction with region-level precision.
- DocTags** – Minimal, efficient, and fully compatible with DoclingDocuments.
- Layout Awareness** – Preserves structure, positions, and element bounding boxes.
- Code, Formula, Figures** – Recognizes complex structured content.
- Charts & Tables** – Parses and reconstructs structured data visually.
- Captions, Lists, & Headers** – Maintains hierarchy and semantic links.
- Full-Page Conversion** – Converts complete pages with all content types.
- General Document Support** – Trained on diverse document types.
- Fast Inference** – ~0.35 sec/page on A100 with VLLM.

Scale, cost and performance



Leandro von Werra • 1st
Head of Research at Hugging Face
15h •

The FinePDFs pipeline really shows the massive scale needed for LLM data processing (spoiler: it's ~\$1M):

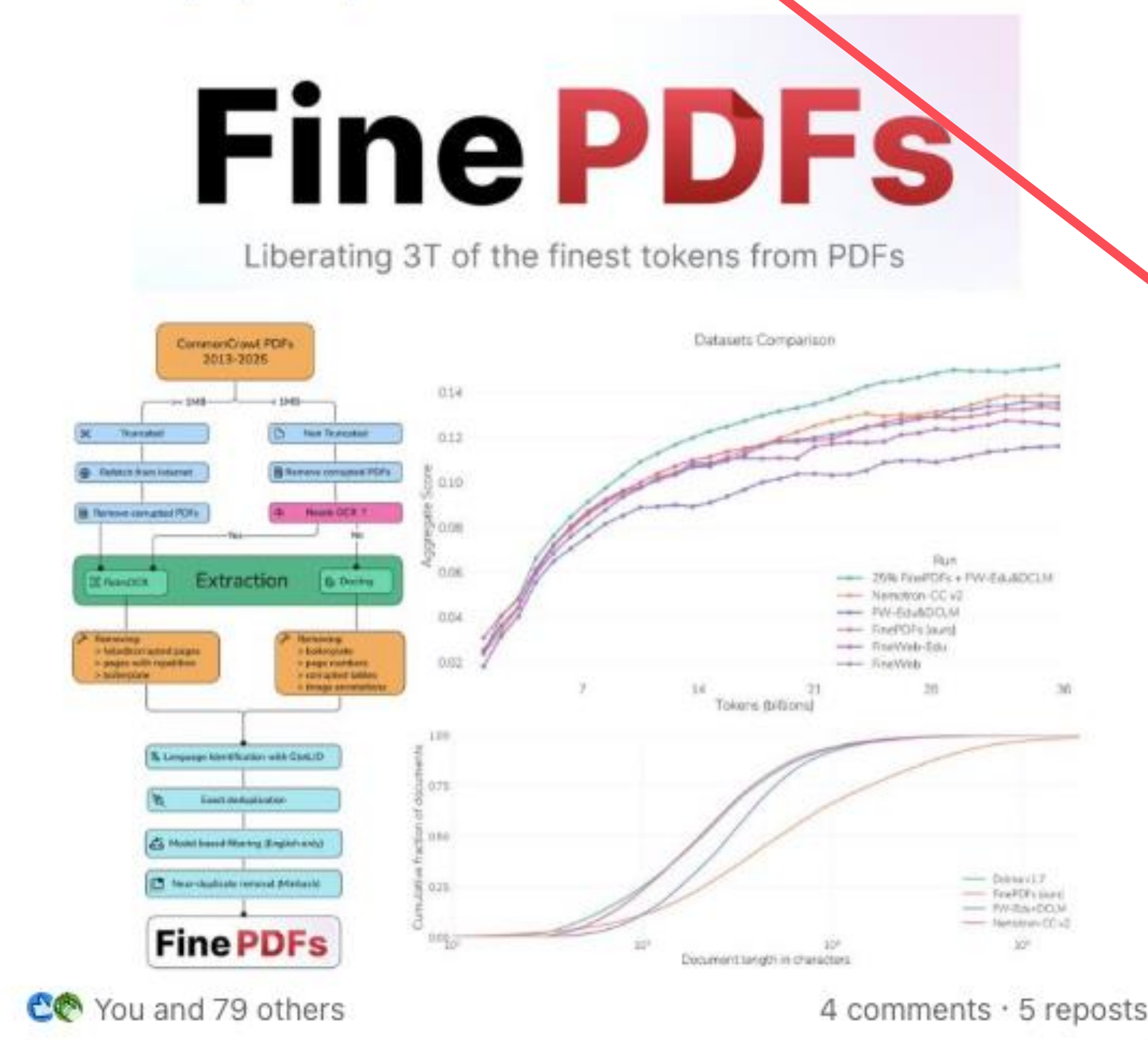
Raw data: 1.35B PDFs (1.2 PB) → \$50k/mo storage

Extraction:
> rolmOCR: 368M docs → 250k GPUh, \$750k
> docling: 918M docs → 2.4M vCPUh, \$35k

Total liberation cost (incl. ablations): ~\$1M

Dataset: <https://lnkd.in/e-muAYtJ>

- 📖 475,019,140 PDFs parsed end-to-end
- 🌐 1,733 languages represented
- 📖 ~3 trillion tokens (~2,918B) extracted
- 📁 3.65 TB of high-quality, deduplicated text
- 📅 Data spanning 2013–2025 across 105 CommonCrawl snapshots



→ $918/368 * 750/35 = 50$

→ Docling is 50 times more cost-effective than VLMs!!

Creando un Sistema RAG Multimodal con Docling



Workshop

- Conversión de Documentos con Docling
- Chunking Avanzado con Docling
- Implementamos un RAG Multimodal con Docling



docling-workshop-PyConES-2025

 [ibm.biz/pycones25](https://github.com/ibm.biz/pycones25)

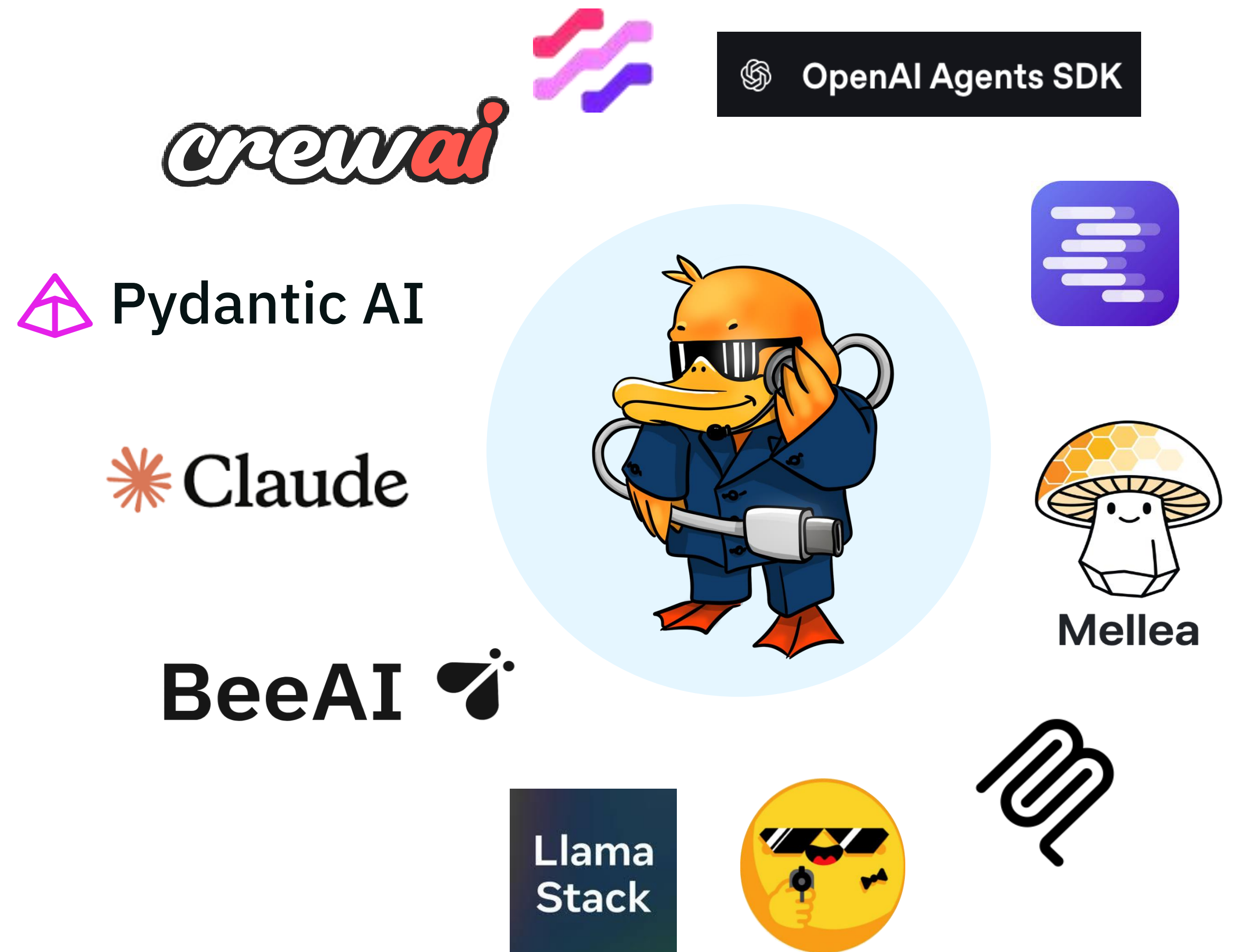
What is next?

Make Docling agentic!

- An AI agent is an application that can **act autonomously** to understand, plan, and execute a specific task.
- MCP is a **protocol** for connecting AI applications to external tools and data sources.
- **Docling** provides an MCP server with tools for document conversion, manipulation, or content creation
- Integration examples with several frameworks and AI desktop clients
- Docling MCP in action with LM Studio with a single click



Add to LM Studio



What is next?

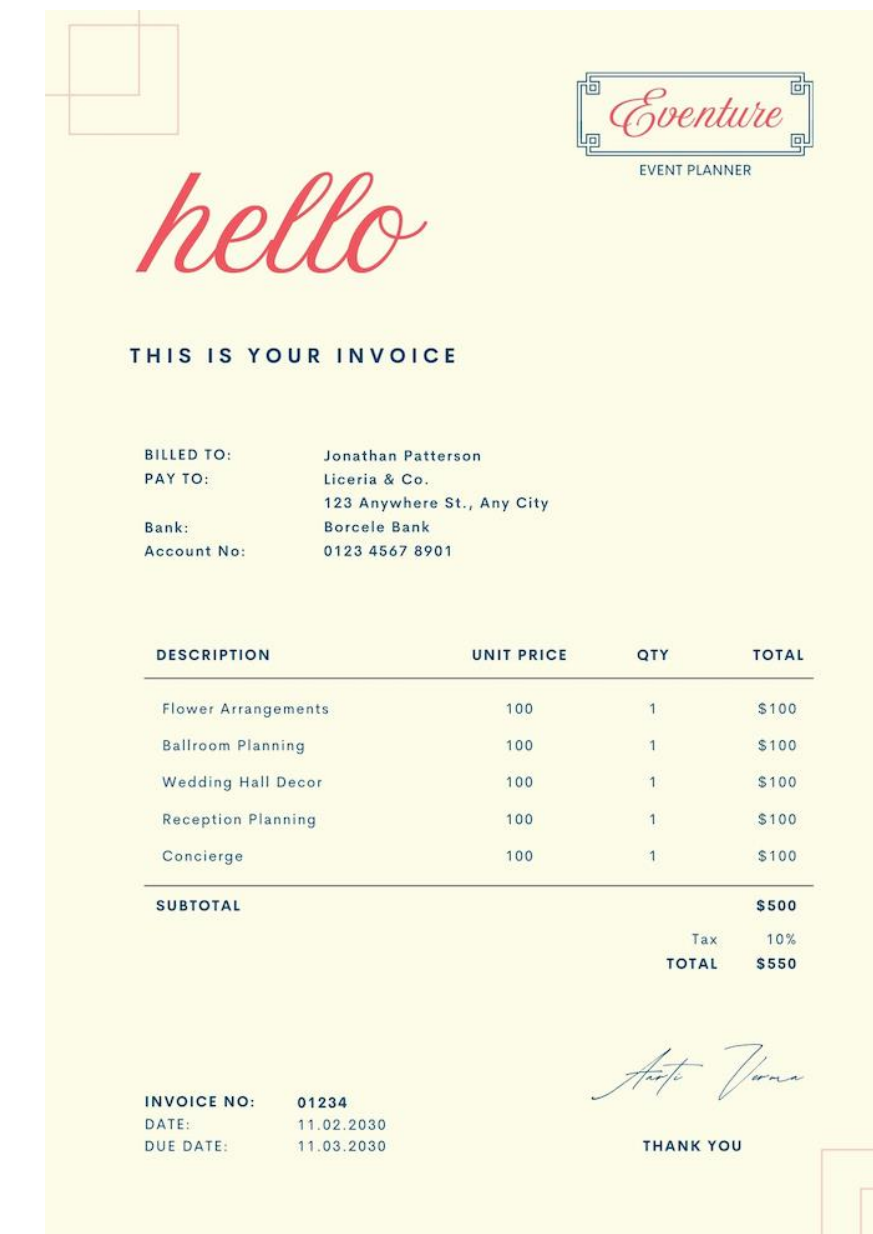
Schema based data extraction

```
from docling.datamodel.base_models import InputFormat
from docling.document_extractor import DocumentExtractor

formats=[InputFormat.IMAGE, InputFormat.PDF]
extractor = DocumentExtractor(allowed_formats=formats)

invoice_dict = {
    "bill number": "string",
    "total invoice price": "float",
    "currency of total invoice price": "string",
    "name of invoice addressee": "string",
    "name of invoice sender": "string",
}

result = extractor.extract("invoice.jpg", template=invoice_dict)
print(result)
```



1

Define the desired extraction schema

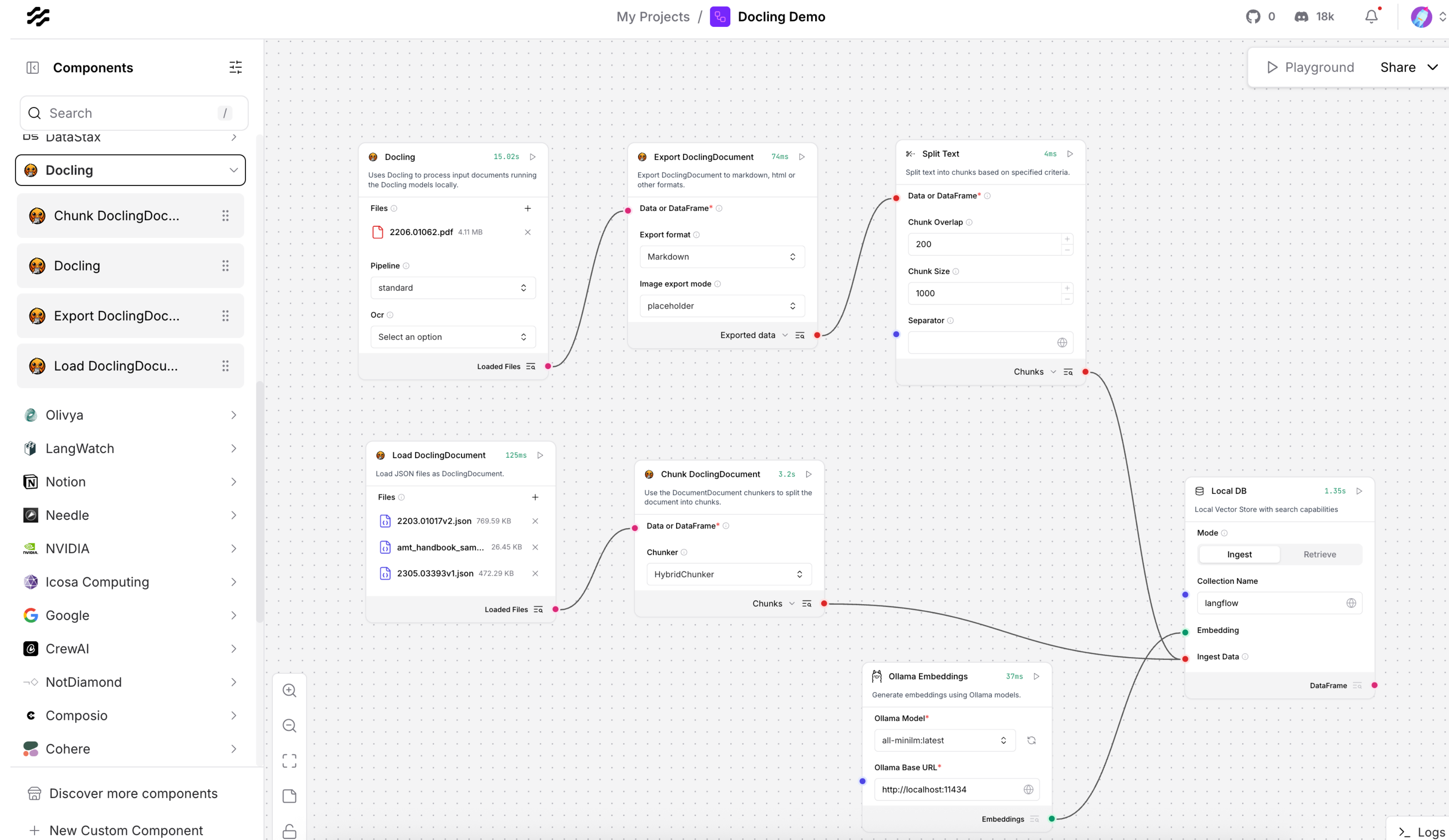
2

Run the Docling DocumentExtractor

```
{
    'bill number': '01234',
    'total invoice price': 550,
    'currency of total invoice price': 'USD',
    'name of invoice addressee': 'Jonathan Patterson',
    'name of invoice sender': 'Eventure Event Planner'
}
```

What is next?

No-Code, Low-Code pipelines



- Convert documents with Docling
- DoclingDocument as supported type in the flow
- Export to text / markdown for common flows
- Native Docling chunker

Docling

Get your documents ready for gen AI



Workshop

- Try out Docling
- Build a multimodal RAG application with Docling
- Develop agentic AI use cases

Try it out!

🔗 GitHub: github.com/docling-project

🌐 Web: docling.ai

📦 Install: `pip install docling`

📖 Tech paper: arxiv.org/abs/2501.17887



💻 Use it, ★ star it,
🚀 spread the word about Docling!

